

MATERIALS MANAGEMENT – NOTES

1. PROCUREMENT

Procurement is the process of acquiring materials in the right quantity, right quality, at the right time and at minimum cost. It includes planning, supplier selection, purchasing, and follow-up to ensure uninterrupted production.

Purchase Procedure

The purchase procedure starts with a purchase requisition, followed by enquiry to suppliers, receipt and comparison of quotations, selection of supplier, issue of purchase order, follow-up, and receipt of materials.

User Department



Purchase Requisition



Enquiry to Suppliers



Receipt of Quotations



Comparison of Quotations



Supplier Selection



Purchase Order



Follow-up



Receipt of Materials

Tender

A tender is a formal invitation to suppliers to submit quotations. It may be open, limited, or single.

Need Identified



Tender Notice



Submission of Tenders



Opening of Tenders



Technical Evaluation



Financial Evaluation



Selection of Supplier

Earnest Money Deposit (EMD)

EMD is a small amount paid along with tender to show seriousness of the bidder and is refunded to unsuccessful bidders.

Tender Stage



EMD Paid by Bidders



Supplier Selected



Security Deposit Paid



Order Execution



Successful Completion



Security Deposit Refunded

Security Deposit

Security deposit is collected from the selected supplier to ensure proper execution of the contract.

Materials Received



Invoice Verification



Quantity Check



Quality Inspection



Accepted / Rejected



Store Entry

Purchase Order

A purchase order is a legal document specifying material details, quantity, price, and delivery terms.

Vendor Rating

Vendor rating evaluates suppliers based on quality, delivery performance, price, and service.

2. RECEIPT OF MATERIALS

Receipt of materials involves verifying and accepting materials supplied by vendors.

Invoice

Invoice shows material details, price, taxes, and total amount payable.

Cash Memo

Cash memo is issued when immediate payment is made for small purchases.

Inspection

Inspection ensures that materials received meet quantity and quality requirements.

Materials Received

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Invoice Verification

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Quantity Check

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Quality Inspection

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Accepted / Rejected

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Store Entry

3. STORAGE

Storage refers to safe keeping of materials until required for use.

Methods of Storage

Centralized and decentralized storage systems are used. Open storage is used for bulky materials, closed storage for valuable items, and FIFO method avoids deterioration.

Receipt of Materials



Classification



Proper Storage Method



Preservation & Protection



Issue to Departments

4. SELECTIVE INVENTORY CONTROL TECHNIQUES

Selective control focuses attention on important inventory items.

Old Stock → Used First

New Stock → Used Later

ABC Analysis

ABC analysis classifies items based on annual consumption value into A, B, and C categories.

Inventory Items



Annual Consumption Value



Classification

A B C(High) (Medium) (Low)

Category Items Value Control

Category	Items	Value	Control
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A	Few	High	Strict
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B	Moderate	Medium	Moderate
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C	Many	Low	Simple
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Case study:

A manufacturing company stores the following items in its store.

The materials manager wants to apply ABC analysis to decide which items need strict control.

Item	Annual Usage (units)	Unit Cost (₹)
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A	100	500
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B	200	200
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C	1,000	20
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D	2,000	5
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E	5,000	1
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STEP 1: Calculate Annual Consumption Value

Annual Consumption Value = Annual Usage × Unit Cost

Item	Usage	Cost (₹)	Annual Value (₹)
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A	100	500	50,000
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B	200	200	40,000
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C	1,000	20	20,000
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Item Usage Cost (₹) Annual Value (₹)

D 2,000 5 10,000

E 5,000 1 5,000

STEP 2: Arrange Items in Descending Order of Value

Rank Item Annual Value (₹)

1 A 50,000

2 B 40,000

3 C 20,000

4 D 10,000

5 E 5,000

STEP 3: Calculate Total Consumption Value

Total = 50,000 + 40,000 + 20,000 + 10,000 + 5,000

= ₹1,25,000

STEP 4: Find Percentage Contribution

Item Value (₹) % of Total

A 50,000 40%

B 40,000 32%

C 20,000 16%

Item	Value (₹)	% of Total
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D	10,000	8%
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E	5,000	4%
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STEP 5: Classify into A, B, and C Categories

A items → Top **70% value** (strict control)

B items → Next **20% value**

C items → Last **10% value**

✓ FINAL ABC CLASSIFICATION RESULT

Category	Items	Control Required
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A	A, B	Very strict control
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B	C	Moderate control
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C	D, E	Simple control
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VED Analysis

VED analysis classifies items based on criticality as Vital, Essential, or Desirable.

Inventory Items

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Criticality

V E D

Vital Essential Desirable

5. INVENTORY THEORY

Inventory theory deals with deciding how much to order and when to order.

Demand Occurs



Check Stock Level



Reorder Level Reached?



Yes → Place Order



Lead Time



Stock Received

Objectives of Keeping Inventory

Inventory ensures smooth production, avoids stock-outs, meets sudden demand, and acts as a buffer.

Structure of Inventory Problems

Inventory problems involve demand rate, lead time, reorder level, and costs.

Relevant Costs

Relevant costs include ordering cost, carrying cost, purchase cost, and shortage cost.